

Interpreting the Solution of a Linear System on a Graph

Answer Key

Algebra 1

Quick Answers

1. $x = 2, y = 3$

2. $x = 3, y = 2$

3. $x = 2, y = 0$

4. $x = 1, y = 3$

5. —

6. $x = 5, y = 40$

7. $x = 1, y = 4$

8. $x = 2, y = 2$

9. $x = -1, y = 2$

10. —

Curriculum

Level: Algebra 1

HSA-REI.C.6 – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–4 of 5 across 10 tagged checks.

1. $y = x + 1$ and $y = -x + 5$ — **read the crossing point.**

Reading the graph. ℓ_1 climbs one square up for each square right, ℓ_2 falls one for each square right, and they cross at the lattice point two right and three up.

Checking, which is the part that matters. A crossing point read by eye is a claim, not a proof, so substitute into *both* original equations: $3 = 2 + 1$ ✓ and $3 = -2 + 5$ ✓. Both are true, so the point really is on both lines. $\boxed{(2, 3)}$

2. $y = 2x - 4$ and $y = -x + 5$ — **read the crossing point.**

Reading the graph. ℓ_1 starts at -4 on the y -axis and rises two squares per square right; ℓ_2 starts at 5 and falls one per square right. They meet three right and two up.

Checking. $2 = 2(3) - 4$ ✓ and $2 = -3 + 5$ ✓. $\boxed{(3, 2)}$

3. Graph $y = x - 2$ and $y = -2x + 4$.

Graphing. Both are already in slope-intercept form. For $y = x - 2$ plot $(0, -2)$ and step up 1, right 1. For $y = -2x + 4$ plot $(0, 4)$ and step down 2, right 1, which lands on $(1, 2)$ and then $(2, 0)$.

Reading and checking. The lines cross on the x -axis, at two right and zero up. Substituting: $0 = 2 - 2$ ✓ and $0 = -2(2) + 4$ ✓. A solution with $y = 0$ is not a special case — it just means the crossing point happens to sit on the x -axis. $\boxed{(2, 0)}$

4. $x + y = 4$ and $y = 2x + 1$.

Reading the graph. ℓ_1 is $x + y = 4$, the line through $(0, 4)$ and $(4, 0)$; it was never rearranged, and it did not need to be. ℓ_2 starts at $(0, 1)$ and rises two per square right. They cross one right and three up.

Checking in the original equations. $1 + 3 = 4$ ✓ and $3 = 2(1) + 1$ ✓. Checking in the *original* forms is the point here: if a student rearranges $x + y = 4$ incorrectly, checking against the rearranged version would hide the error. $(1, 3)$

5. $y = 2x + 1$ and $y = 2x - 3$ — **parallel lines.**

What the graph shows. Both lines rise two squares for every square right, so they never get closer together; the second is simply the first slid down 4. They never touch, so no point lies on both.

Confirming it algebraically. A crossing point would need $2x + 1 = 2x - 3$. Subtracting $2x$ from both sides leaves $1 = -3$, which is false for every x , so there is no such point.

no solution — the lines are parallel

Grading: the explanation should say “same slope, different y -intercept” or “they never meet”, not just “zero”.

6. Break-even: $y = 15 + 5x$ and $y = 8x$.

Graphing. Harbor Rentals starts at $(0, 15)$ — the launch fee is paid before any hours — and rises 5 dollars per hour. Bayside starts at the origin and rises 8 dollars per hour. Bayside is steeper, so it starts below and catches up.

Reading and checking. The lines cross at five hours and 40 dollars: $15 + 5(5) = 40$ ✓ and $8(5) = 40$ ✓.

What it means. At exactly 5 hours the two shops cost the same, 40 dollars. For fewer than 5 hours Bayside is cheaper (its line is lower); for more than 5 hours Harbor Rentals is cheaper. That reading of “which line is lower” is the interpretation the question asks for — the point itself is only where the answer switches. $(5, 40)$

7. $2x + y = 6$ and $4x + 2y = 12$ — **one line drawn twice.**

What the graph shows. Solving each for y gives $y = -2x + 6$ and $y = -2x + 6$: the same slope and the same intercept. The second equation is the first multiplied by 2, so plotting it retraces the first line exactly.

How many solutions. Every point on that line satisfies both equations, so there are infinitely many. Two easy ones: $(1, 4)$ since $2 + 4 = 6$, and $(2, 2)$ since $4 + 2 = 6$. The full solution set is all ordered pairs with $y = -2x + 6$.

infinitely many solutions; e.g. $(1, 4)$ and $(2, 2)$

Note for the grader: the verifier confirms the listed point lies on both lines and reports the system as an infinite family for human confirmation — a solution set with infinitely many members cannot be enumerated by machine.

8. $y = \frac{1}{2}x + 1$ and $y = -x + 4$.

Graphing. The first starts at $(0, 1)$ and rises one square for every *two* squares right, so plot $(0, 1)$, then $(2, 2)$, then $(4, 3)$. The second starts at $(0, 4)$ and falls one per square right.

Reading and checking. They cross two right and two up. Substituting: $\frac{1}{2}(2) + 1 = 2$ ✓ and $-2 + 4 = 2$ ✓. A fractional slope is where reading a graph by eye goes wrong most often, which is why the check is not optional. $\boxed{(2, 2)}$

9. Find the mistake: $y = -x + 1$ and $y = 3x + 5$.

Testing the student's point. Substitute $(2, -1)$ into the first equation: $-2 + 1 = -1$ ✓ — it does lie on ℓ_1 . Now the second: $3(2) + 5 = 11$, not -1 ×. A point must satisfy *both* equations, so $(2, -1)$ is not the solution.

The correct solution. The lines cross one left and two up. Check: $-(-1) + 1 = 2$ ✓ and $3(-1) + 5 = 2$ ✓.

What the student did. They read the crossing point and wrote the coordinates in the wrong order and with the wrong signs — reading up-then-across instead of across-then-up. Ordered pairs are always (across, up). $\boxed{(-1, 2)}$

10. Explain: why never exactly two solutions?

Expected reasoning (grade by hand). Two distinct points determine exactly one line. If two lines shared two different points, then each line would be the one line through those two points, so the lines would be the same line — and then they share not two points but every point. So sharing exactly two is impossible.

That leaves three cases, and the graph shows each one: the lines cross once (one solution), they are parallel and never cross (no solution), or they are the same line (infinitely many).

Grading: full credit for any argument that two shared points force the lines to coincide, or for a clear case-by-case argument on slopes: equal slopes with different intercepts give none, equal slopes with equal intercepts give infinitely many, unequal slopes give exactly one. Sketches alone, without the reason, are partial credit.